

Amirhossein Masoudnezhad

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Profile

Master's student in Astrophysics and Cosmology interested in observational cosmology and cosmological simulations. Experienced in Python-based scientific computing, numerical methods, and HPC environments, with hands-on work in astronomical data analysis and large-scale numerical simulations. Background in multi-wavelength observations and machine-learning techniques provides a flexible toolkit for tackling complex cosmological problems.

Education

M.Sc. in Astrophysics & Cosmology

2024–Present

University of Bologna

Relevant Coursework: Multi-Wavelength Astrophysics Laboratory, Practical Statistics, High-Performance Computing, Advanced Cosmology

B.Sc. in Physics

2019–2022

Sharif University of Technology

Relevant Coursework: Cosmology, Quantum Mechanics, Mathematical Physics, Machine Learning

B.Sc. in Electrical Engineering

2016–2019

Sharif University of Technology

Relevant Coursework: Signals and Systems, Communication Systems, Electronics

Research Interests

- Observational Cosmology and Cosmological Simulations
- Early-Universe Physics and Perturbations
- Scientific Software Development and Numerical Methods

Publications

- **Dusty stellar sources classification by implementing machine learning methods based on spectroscopic observations in the Magellanic Clouds.** The Astrophysical Journal, 2025
- **Spectral identification and classification of dusty stellar sources using spectroscopic and multiwavelength observations through machine learning.** IAU Symposium 368, 2023
- **Machine Learning Approach for the Spectral Classification of Dusty Stellar Sources in the Magellanic Clouds.** EAS Publications Series, 2024

Research Experience

Optical Spectroscopy of Globular Cluster Giant Stars

- Reduced and normalized high-resolution optical spectra
- Measured radial velocities and derived stellar abundances
- Validated results against reference cluster values using SLAB and SALVADOR

Observation Planning for Bootes II RGB Stars

- Designed an observing strategy for RGB stars in an ultra-faint dwarf galaxy
- Optimized telescope/instrument selection, exposure times, and observing conditions

- Achieved target signal-to-noise ratio and spectral resolution

Calibrating and Imaging VLA Data of RBS797

- Performed full calibration and imaging of VLA continuum data using CASA
- Produced cleaned radio images and measured flux densities and spectral indices
- Estimated physical properties of the central radio source

Fermi-LAT Analysis of TXS 0506+056

- Processed five years of gamma-ray data using Fermipy
- Generated counts maps, light curves, and spectral energy distributions
- Analyzed time variability and multi-wavelength emission behavior

Machine-Learning Classification of Variable Stars

- Developed automated pipelines for light-curve analysis
- Implemented classification algorithms for variable star types

Strong Scaling Test of Gadget-2 N-Body Simulations

- Ran cosmological N-body simulations on multi-core HPC clusters
- Evaluated strong-scaling performance and parallel efficiency

Technical Skills

Scientific Programming & Data Analysis

Python (NumPy, SciPy, Astropy), C/C++, Fortran, MATLAB, Bash scripting, Git, Linux/HPC environments, MPI (HPC)

Astronomical Data & Observations

Optical Spectroscopy (SLAB, SALVADOR)
Radio Astronomy (CASA)
Gamma-ray Analysis (Fermipy, Fermi-LAT)
Observation planning, calibration, time-series analysis

Machine Learning

Supervised and unsupervised learning, neural networks, deep learning

Additional Technical Skills

Web & Back-end Development (PHP, SQL, MongoDB)
App Development (Flutter/Dart, Swift)

Honors & Awards

- Ranked **103** among 160,000+ participants in National University Entrance Exam (Math track)
- Silver Medal – National Astronomy & Astrophysics Olympiad
- Bronze Medal – National Astronomy & Astrophysics Olympiad
- Member – National Elites Foundation
- Financial scholarship for academic excellence (INEF)
- Member – Young Scholars' Club

Additional Activities

Organizing Committee Member – National Observatory Conference (2022)

Olympiad Instructor – Astronomy & Astrophysics, Salam High School (2017)

Languages

English: Fluent (TOEFL iBT 96)